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1. Presence of elongated rod like notochord below the nerve cord and above the alimentary canal. It serves as a primitive internal skeleton. It may persist throughout life in lancelets and lampreys. In adult vertebrates, it may be partially or completely replaced by backbone or vertebral column.
2. A dorsal hollow or tubular fluid filled nerve cord lies above the notochord and below the dorsal body wall. It serves to integrate and co-ordinate the body functions. In higher chordates, the anterior end of the nerve cord gets enlarged to form the brain and the posterior part becomes the spinal cord, protected inside the vertebral column.
3. Presence of pharyngeal gill slits or clefts in all chordates at some stage of their lifecycle. It is a series of gill slits or clefts that perforates the walls of pharynx and appears during the development of every chordate. In aquatic forms, pharyngeal gill slits are vascular, lamellar and form the gills for respiration. In terrestrial chordates, traces of non-functional gill clefts appear during embryonic developmental stages and disappear later. Besides the above said features, chordates are bilaterally symmetrical, triploblastic, coelomates with organ system level of organisation; they possess post anal tail, closed circulatory system with a ventral myogenic heart except in *Amphioxus*.

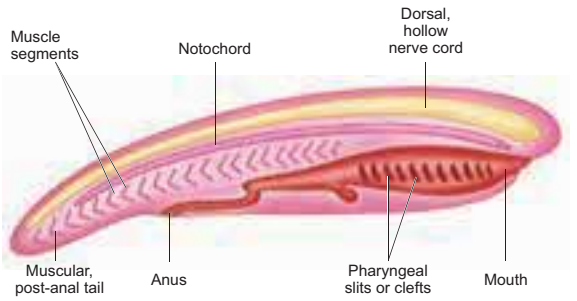


Figure 2.19 A Typical Chordate

2.4.1. Subphylum: Urochordata or Tunicata

(G. *Oura* – A tail; L. *Chord* – cord)

They are exclusively marine and are commonly called sea squirts. Mostly sessile, some pelagic or free swimming, exist as solitary and colonial forms. Body is unsegmented and covered by a test or tunic. Adult forms are sac like. Coelom is absent, but has an atrial cavity surrounding

Table. 1. Comparison of chordates and non-chordates

Chordates	Non-chordates
Notochord is present	Absence of notochord
Dorsal, hollow and single nerve cord	Double ventral solid nerve cord
Pharynx perforated by gill slits	Gill slits absent
Heart is ventrally placed	Heart is dorsal or laterally placed or absent
A post anal tail is present	Post anal tail is absent
Alimentary canal placed ventral to the nerve cord	Alimentary canal is placed dorsal to the nerve cord

List the three features common to all chordates at sometime in their life.